MC300 Summer Training

Report submitted in partial fulfillment of the requirements for the award of a degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING
(VI SEMESTER)

BY

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JAIPUR CAMPUS, JAIPUR SP-2024

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July 08, 2024

APPROVAL OF THE PROJECT

This is to certify that the work presented in the project entitled **" *Smart Vehicles Surveillance in Foggy Conditions using Enhanced Deep Learning Algorithms* "** is submitted by ***Khush Khandelwal (BTECH/25028/21)*** and ***Shubham Kumar Sain (BTECH/25069/21)*** under the guidance of ***Dr. Madan Mohan Agarwal*** in partial fulfilment of the requirement for the award of Degree of Bachelor of Computer Application in Computer Science and Engineering of Birla Institute of Technology, Mesra, Ranchi, Extension Center Jaipur is an authentic work carried out under supervision and guidance of us.

To the best of our knowledge, the content of this project does not form a basis for the award of any previous degree to anyone else.

Dr. Madan Mohan Agarwal
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ACKNOWLEDGEMENT

I would like to express my deepest gratitude to all those who have contributed to the successful completion of the project titled "Smart Vehicles Surveillance in Foggy Conditions using Enhanced Deep Learning Algorithms."

First and foremost, I am profoundly grateful to Mr. Madan Mohan Agarwal for his invaluable guidance, support, and encouragement throughout this project. His expertise and insights have been instrumental in steering this project to its successful completion. His unwavering commitment to excellence has been a constant source of inspiration.

We perceive this opportunity as a great opportunity in my career development. I will strive to use the skills and knowledge we gained during my time creating this project. It will help me in my future work.

Thank you all for your contributions, which have been vital in the successful completion of this project.

Sincerely,

Khush Khandelwal BTECH/25028/21

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ABSTRACT

Foggy conditions pose significant challenges for vehicles due to reduced visibility, making it difficult for drivers to see other vehicles, pedestrians, and road signs, which increases the risk of accidents. Fog also distorts depth perception, complicating the ability to accurately judge distances and speeds of other vehicles, leading to potential collision. For this we use deep learning methods, while deep convolutional neural networks excel at eliminating fog, they also need to be able to handle photos taken in actual meteorological situations with patches of cloud cover or fog. Blur is harder to categorize in the actual world, and decreasing map or picture quality will result in output results with inconsistent colors or less content. Additionally, the model's complexity will rise with additional convolutional block stacking. Deep learning methods for fog image processing can be plagued by overfitting in addition to the challenge of gathering enough training data. This can restrict the capabilities of the model and make it difficult to use it practically in real- world situations.

This project is a combined method for removing fog from surveillance images using Wavelet Former Net, a Transformer-based wavelet network designed for real-world non-homogeneous dense fog scenarios, this transformer method uses the wavelet transform method. It also uses Multi-Object Detection with Enhanced YOLOv2 and LuNet Algorithms to detect objects. When these methods are combined, they can better handle the intricacies of hazy surroundings, improving both visibility and object detection precision. The effectiveness of the proposed technique is proven through rigorous testing, highlighting its potential to enhance the operation of monitoring systems in difficult weather situations.

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Introduction of the Project and Methods Smart Vehicles Surveillance in Foggy Conditions using Enhanced Deep Learning Algorithms

Foggy conditions pose significant challenges for vehicles due to reduced visibility, making it difficult for drivers to see other vehicles, pedestrians, and road signs, which increases the risk of accidents. Fog also distorts depth perception, complicating the ability to accurately judge distances and speeds of other vehicles, leading to potential collision. For this we use deep learning methods, while deep convolutional neural networks excel at eliminating fog, they also need to be able to handle photos taken in actual meteorological situations with patches of cloud cover or fog. Blur is harder to categorize in the actual world, and decreasing map or picture quality will result in output results with inconsistent colors or less content. Additionally, the model's complexity will rise with additional convolutional block stacking. Deep learning methods for fog image processing can be plagued by overfitting in addition to the challenge of gathering enough training data. This can restrict the capabilities of the model and make it difficult to use it practically in real- world situations.

This proposed a combined method for removing fog from surveillance images using Wavelet Former Net, a Transformer-based wavelet network designed for real-world non-homogeneous dense fog scenarios, this transformer method uses the wavelet transform method. It also uses Multi-Object Detection with Enhanced YOLOv2 and LuNet Algorithms to detect objects. When these methods are combined, they can better handle the intricacies of hazy surroundings, improving both visibility and object detection precision. The effectiveness of the proposed technique is proven through rigorous testing, highlighting its potential to enhance the operation of monitoring systems in difficult weather situations.

Wavelet Former Net method

Transformer-based models have recently produced good results for dehazing photos. The WaveletFormer and IWaveletFormer blocks are used in this process to prevent texture detail loss and preserve image quality. The lightweight mechanism's multi- frequency information is captured by the parallel convolution in the Transformer blocks. We provide a feature aggregation module (FAM) to further improve WaveletFormerNet's feature extraction capabilities by capturing long-range relationships among data of various levels. We introduce an end-to-end wavelet reconstruction system called WaveletFormerNet.

By introducing the WaveletFormer and IWaveletFormer blocks, this novel technique minimizes the loss of texture detail while maintaining picture quality. WaveletFormerNet is a lightweight technique that effectively gathers multi-frequency information by utilizing parallel convolution inside Transformer blocks. Furthermore, to improve feature extraction, a feature aggregation module (FAM) is presented, which captures long-range relationships among data at several levels.

Enhanced YOLOv2 and LuNet

The use of LuNet and Enhanced YOLOv2 algorithms in surveillance videos is suggested by T. Mohandoss and J. Rangaraj. In the Yolov2 algorithm. In this study, a real-time video dataset known as MOT20 was collected and converted into minuscule video frames. The video/image frame of the recognized object will provide recommended layouts and noise detection for different moving objects. The noise is eliminated and smoothed with a Kalman filter. The filter makes use of noise measurements gathered over time to anticipate future observations and assess the model's parameters. Forecasts, measurements, and updates based on forecasts and comparisons at every level are all possible with this filter. Mathematical estimators may be used to anticipate and update the state of various linear processes. In order to improve performance, the YOLOv2 network predicts and categorizes bounding box categories using binary cross-entropy loss rather than multiple labels. This effort replaced the number of metrics in the YOLOv2 base network with LuNet, so altering it. This study uses LuNet technology for feature extraction in the upgraded model to extract the most anticipated qualities from the picture. Additionally, the LuNet design of the underlying network makes the solution compact. This study employs LuNet as the base network, which enables us to harvest important features from the first layer and transmit them to the last layer due to the direct connections between all layers.

Clear Detect

YOLO (You Only Look Once) object detection, like many other computer vision algorithms, relies heavily on the quality of the input images. Higher-quality images generally provide clearer and more distinguishable visual features, which can aid in the accurate identification and localization of objects. Conversely, low-quality images with noise, blurriness, or other imperfections can hinder the detection process, potentially leading to slower performance and reduced accuracy in predicting object locations and classifications. Therefore, ensuring good quality data inputs is crucial for optimizing the performance of YOLO-based object detection systems.

To address the limitations inherent in individual methods for object detection, we've devised a new approach that merges two distinct techniques. Recognizing that the quality and quantity of images are pivotal factors, we've devised a strategy that leverages the strengths of both methods. Initially, we employed an image fog removal technique to enhance the clarity of the visual data. Subsequently, this refined image is utilized for object detection, thereby augmenting the accuracy of predictions. By combining these methodologies, we aim to achieve superior results by optimizing both image quality and the effectiveness of object detection algorithms.

This paper method is a combined method for removing fog from surveillance images using WaveletFormerNet, a Transformer-based wavelet network designed for real- world non-homogeneous dense fog scenarios; this transformer method uses the wavelet transform method. It also uses Multi-Object Detection with Enhanced YOLOv2 and LuNet Algorithms to detect objects. The combination of these methods known as “ClearDetect”, can better handle the intricacies of hazy surroundings, improving both visibility and object detection precision.

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METHODOLOGY AND WORK DONE

During this project we have found new approaches for creating a better solution and we decided to use yolov2 for its simplicity and easy to work with nature and LuNet algo for better results. Methods have adjusted the algorithms and weights according to our needs. Also used prior experiences of many researchers.

- Creating a combined Algorithm which dehaze the image and detects the objects for surveillance
- Adjusted the dehazing module for better results.
- using YOLOv2 for better object detection.

Technologies used:

- Visual studio code
- Yolov2
- wavelet formernet method
- Github

Languages used:

- Python

WORKING PRINCIPLES

WaveletFormerNet

The WaveletFormerNet's construction is shown in Fig. 1. WaveletFormerNet's encoding and decoding are both based on the WaveletFormer and IWaveletFormer block; however, DWT and IDWT, respectively, are used in place of down sampling and up sampling in the encoding and decoding segments. While the WaveletFormer and IWaveletFormer block serve as the foundational block of the network, primarily integrating the wavelet transformation and Swin Transformer, we don't utilize these two pre- existing tools directly; instead, we enhance them. The wavelet transformation is utilized to convert the features into the frequency domain. Then, the frequency information is employed to direct WaveletFormerNet to retrieve the image's texture and structural details. Additionally, the receptive field brought on by the Swin Transformer is reduced by the parallel convolution. The WaveletFormer and IWaveletFormer block's structure also lessens the details brought on by issues with down sampling loss and other issues. Additionally, we suggest a Feature Aggregation Module (FAM) that combines various information levels to preserve picture resolution and improve the network's receptive field. To extract features in various receptive fields, we lastly modify an atrous spatial pyramid pooling module (ASPP) in the network and adjust dilated convolution with different expansion rates (rate = 3, 6, 9).

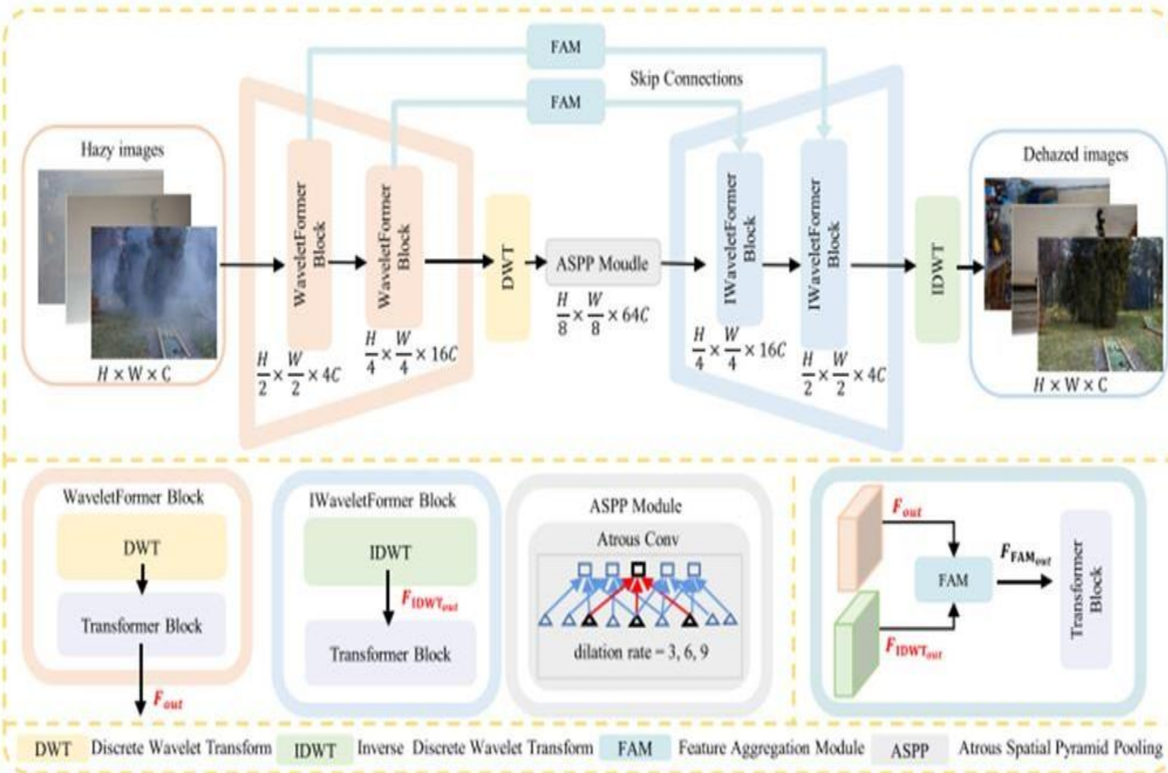


Fig. 1. The schematic illustration of the WaveletFormerNet. WaveletFormer block and IWaveletFormer block consist of DWT and IDWT and Transformer block respectively, and IDWT is the reverse process of DWT.

1. WaveletFormer and IWaveletFormer block

WaveletFormer and IWaveletFormer Blocks use DWT and IDWT to decompose the images from the frequency domain point of view, respectively, and the feature maps are used as inputs to the Transformer module with parallel convolution**.**

2. Frequency decomposition of images

Fig. 1.2 illustrates the detailed structure of the WaveletFormer block, adopting frequency information to guide the network in reconstructing a clear image. We can observe that the input image FDWTin can be divided into the low- and high- frequency details separated into four different frequency subbands: the low- frequency band FLL, the horizontal subband FLH, the vertical subband FHL and the high-frequency subband FHH on the diagonal edge of the original image. This mechanism alleviates detail and color loss and provides a better balance between network processing efficiency and image recovery performance. For the 2D discrete wavelet transform, we import the pytorch_wavelets package and use Daubechies wavelet basis functions.

3. Parallel convolution in Vision Transformer

According to the mechanism , given an input feature map $X \in \mathbb{R}^{b \times h \times w \times c}$, we project X to Q, K, V (query, key, value), and we compute the attention function for a set of queries simultaneously and pack them into a matrix Q; so that the computed output matrix can be described as:

$$\text{Attention}(Q, K, V) = \text{SoftMax}(QK^T/\sqrt{d_k})V \quad \ldots (2)$$

3.1. Feature aggregation module

As the key component of the FAM, the Multi-Head Cross-Attention (MHCA) introduces the feature map Fout of the WaveletFormer block and the feature map FIDWTout from IWaveletFormer block into the MHSA for processing, the computed weight values Y to be rescaled by the sigmoid activation function. The resulting feature tensor Z will be summed with the feature map Fout to obtain the high-level feature tensor. In addition, the feature tensor X produced by FIDWTout is also obtained at a high-level feature tensor after operations such as ReLu. Finally, we concatenate these two high-level feature tensors as the FMHCAout. Therefore, FFAMout can be expressed as:

$$\text{FFAMout} = \text{MLP}(\text{MHCA}(\text{Fout}, \text{FIDWTout})) + \text{Fout} \quad \ldots (3)$$

The feature aggregation module (FAM) introduces the feature map Fout of the WaveletFormer block and the feature map FIDWTout from IWaveletFormer block for processing. fig1.3 illustrates that FAM is a link between the encoding and decoding stages, guiding our WaveletFormerNet to generate images with more crisp textures and rich.

4. Training datasets

We extensively and comprehensively evaluated our model and compared SOTA methods on real-world and synthetic datasets in the same experiment setting.

4.1. Real-world datasets

We use the following four datasets to evaluate our experiments: the NTIRE 2018 image dehazing dataset (I-Haze), the outdoor NTIRE 2018 image dehazing dataset (O-Haze), a benchmark for image dehazing with dense-haze and haze-free images (Dense-Haze), and the NTIRE 2020 dataset for non-homogeneous dehazing challenge (NH-Haze).

4.1.1. I-Haze and O-Haze .

They contain 25 and 35 hazy images (size 2833 × 4657 pixels) respectively for training. Both datasets contain 5 hazy images for validation along with their corresponding ground truth images. We used training data for training and validation data for the test.

4.1.2. Dense-Haze

It contains 45 hazy images (size 1200 × 1600 pixels) for training, 5 hazy images for validation and 5 more for testing with their corresponding ground truth images. We have performed training on training data and tested our model with test data.

4.1.3. NH-Haze

It contains 45 hazy images (size 1200 × 1600 pixels) for training. We selected 40 pairs of data for training and the rest for testing.

4.1.4. Real-world datasets expansion.

We use the same training strategy and dataset expansion process for all four real- world datasets. Specifically, we randomly cropped the original images into square patches of 512 × 512 pixels; these patches are not identical for every epoch. To augment the training data, we implemented random rotations (90, 180, or 270 degrees) and random horizontal flips when processing the training data. This step allows these small real-world datasets to be expanded into larger datasets that are more efficient and more suitable for data-driven methods of training experiments.

Object detection using YOLOv2

It is a development over YOLO. To increase YOLO's speed and detection accuracy, YOLOv2 applies new ideas and incorporates judgments from previous training challenges. YOLOv2 has six stages, which are addressed as follows.

a) **Batch normalization (BN):** Each mini-batch's meaning and variance are calculated and used for activation. After that, the activations are normalized using a zero mean and a one standard deviation for each minibatch. Lastly, the same distribution is used to sample the components of each mini batch. We call this process "batch normalization." It produces the same distribution of activations.

b) **High-resolution classifier:** The input resolution of (224 × 224) is used by the YOLO backbone. In YOLOv2, the input resolution has been improved to (448 × 448).

c) To handle the new resolution input of the object detection job, the network must be adjusted. Consequently, a single-resolution picture (448 × 448) and ten epochs were used to make specific modifications to the classification network in YOLOv2, increasing the average accuracy (mAP) by 1%.

d) **Anchor box convolution:** As previously said, Faster RCNN forecasts bounding boxes by first generating region proposals, which are parameterized with respect to this anchor box. YOLOv2 makes use of this estimating technique. Next, we project the class and object scores for every bounding box prediction. While mAP decreased by 0.3%, withdrawals increased by 7%.

e) **Anchor box size and aspect ratio prediction:** The LuNet method is used by the YOLOv2 to train the bounding boxes to get higher priors. The anchor box's center is then determined by using this backdrop. Using the clustering data, estimate the anchor box's dimensions and aspect ratio. The method improves the accuracy of detection.

f) **Fine-grained features:** As previously stated, YOLO trains with images (224 × 224). The YOLO design has been tweaked to form the Yolov2 architecture. YOLOv2 is retrained using higher resolution images (448 × 448) to pinpoint tiny objects. YOLOv2 uses higher and lower resolution features throughout this retraining process by stacking nearby data in various channels and raises 1% of the detection MAP.

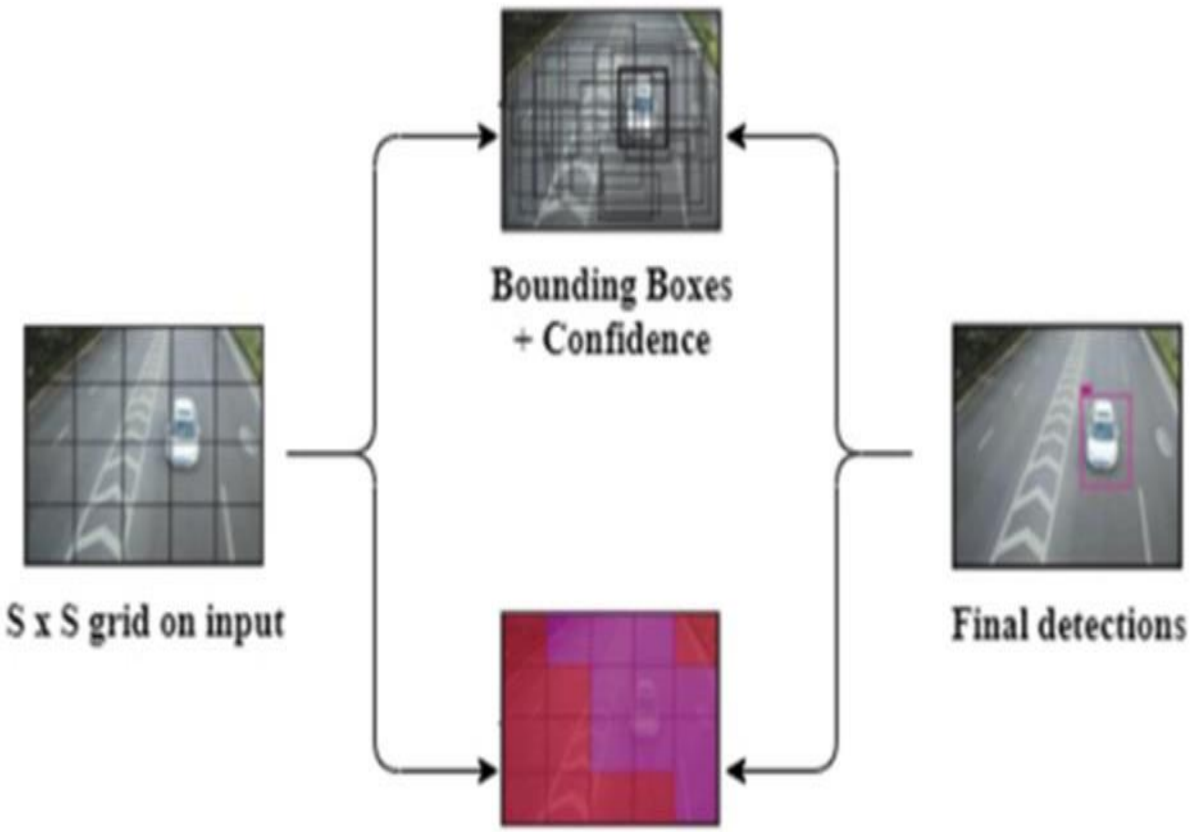


Fig.2. Yolov2 Object detection diagram

IMPLEMENTATION

The hazed original image goes to our dehaze module where it adjusts airtight estimating, calculating boundary constraints, loading filter banks, calculating the transmission map, and finally removing the haze from the image and then it goes to yolov2 code for object detection. Here in fig.3. We have the flow chart of complete process from the waveletformernet Dehazing using wavelet transform methods and then object detection using the yolo and its algorithm

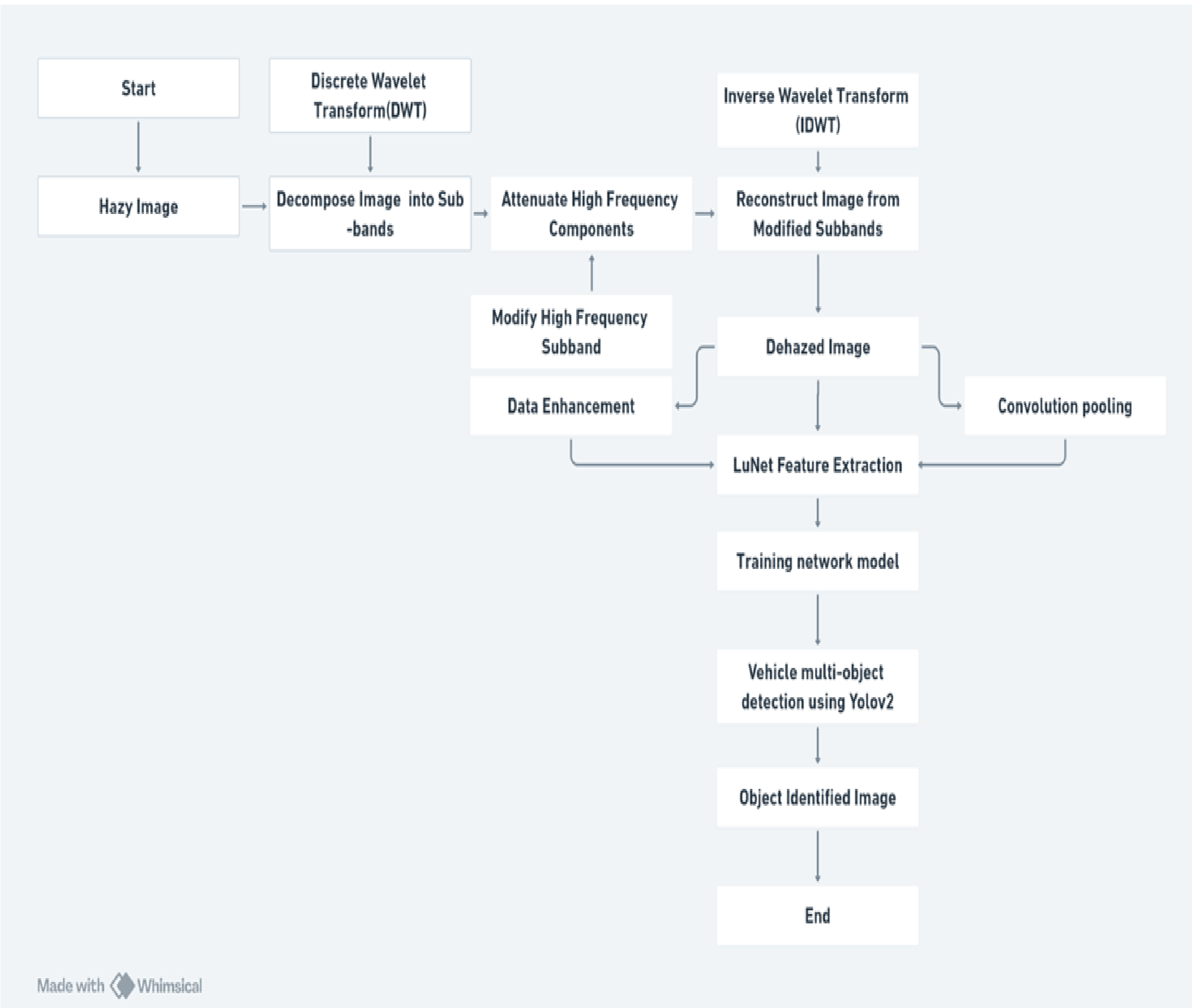


Fig.3. flow chart for the complete process from start to end.

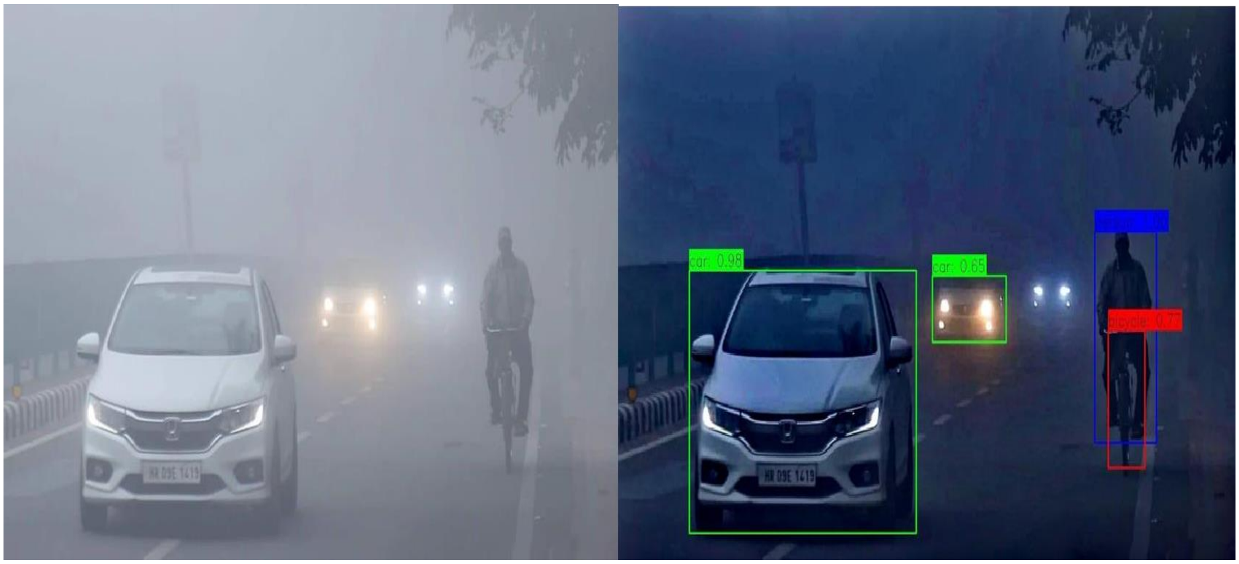


Fig 4.1. First Example image before and after dehazing and object detection

Fig.4.2.1 Second example image before dehazing

Fig.4.2.2. Example Image After Dehazing and object detection. fig.4.2 second example image.

In fig.4.1. and fig.4.2 we have the example images of real-life environment foggy conditions which we have run in our code to get better vision and object detection.

In figure 5.1. and fig.5.2 we have code to implement our dehaze module and then we use

```
def dehaze_image(image_path):
    # Read the image
    frame = cv2.imread(image_path)

    # Check if the image is read successfully
    if frame is None:
        print("Error: Failed to read the image at", image_path)
        return

    # Dehaze the image
    corrected_image = dehaze_frame(frame)

    # Save the dehazed image
    output_path = os.path.join('output', os.path.basename(image_path))
    cv2.imwrite(output_path, corrected_image)
    print(f"Dehazed image saved to {output_path}")

# Function to dehaze frame and perform object detection
def dehaze_frame(frame):
    # Remove haze using the image_dehazer module
    corrected_img, _ = image_dehazer.remove_haze(frame)

    # Perform object detection using YOLO
    net, output_layers, classes = load_yolo_model()
    detected_image = detect_objects_yolo(corrected_img, net, output_layers, classes)

    return detected_image
```

our yolov2 version for object detection.

Fig.5.1. dehazing code for image.

```python
# Load YOLO model
def load_yolo_model():
    yolov3weights = "./yolov3.weights"
    yoloconfig = "./yolov3.cfg"
    net = cv2.dnn.readNet(yolov3weights, yoloconfig)
    with open("./coco.names", "r") as f:
        classes = [line.strip() for line in f.readlines()]
    layer_names = net.getLayerNames()
    unconnected_out_layers = net.getUnconnectedOutLayers()

    # Adjusted output layers indexing
    output_layers = [layer_names[i - 1] for i in unconnected_out_layers.flatten()]
    return net, output_layers, classes

if __name__ == "__main__":
    input_directory = "images"
    output_directory = "output"
    os.makedirs(output_directory, exist_ok=True)

    for filename in os.listdir(input_directory):
        if filename.endswith((".jpg", ".jpeg", ".png")):
            input_image_path = os.path.join(input_directory, filename)
            dehaze_image(input_image_path)
```

fig.5.2. object detection code for image.

Results and Discussion

An image dataset for traffic analysis is used, proposed method “clear detect" clear the foggy image using wavelet transform and YOLOv2-LuNet processing is depicted in . The metrics examined are accuracy, precision, Recall, Detection rate (DET), true positive rate (TP).

The proposed method’s accuracy values are evaluated against current methods by employing feature masking in image. The existing technologies’ accuracy ratings for FFNN, CNN, RCNN, LSTM, and RNN-LSTM

1. Accuracy

The existing technologies’ accuracy ratings for FFNN, CNN, RCNN, LSTM, and RNN-LSTM are 75%, 80%, 84%, 88%, and 88%, respectively. An accuracy of 94% was observed in the case of the proposed model. Figure depicts that the proposed method has a 94% maximum accuracy value**.**

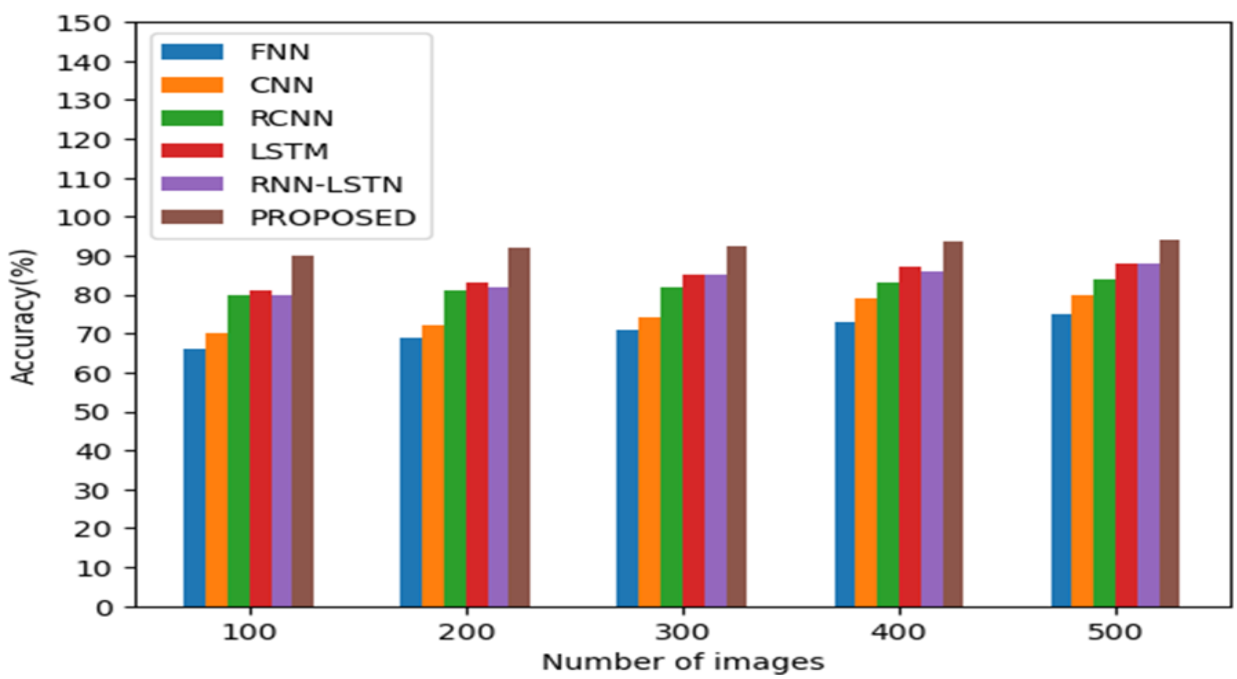


Fig.6. Accuracy of the Methods

2. Precision

The existing technologies' precision ratings for FFNN, CNN, RCNN, LSTM, and RNN-LSTM are 70%, 74%, 85%, 83%, and 83%, respectively. Fig. depicts that the proposed method has a 96% maximum precision value.

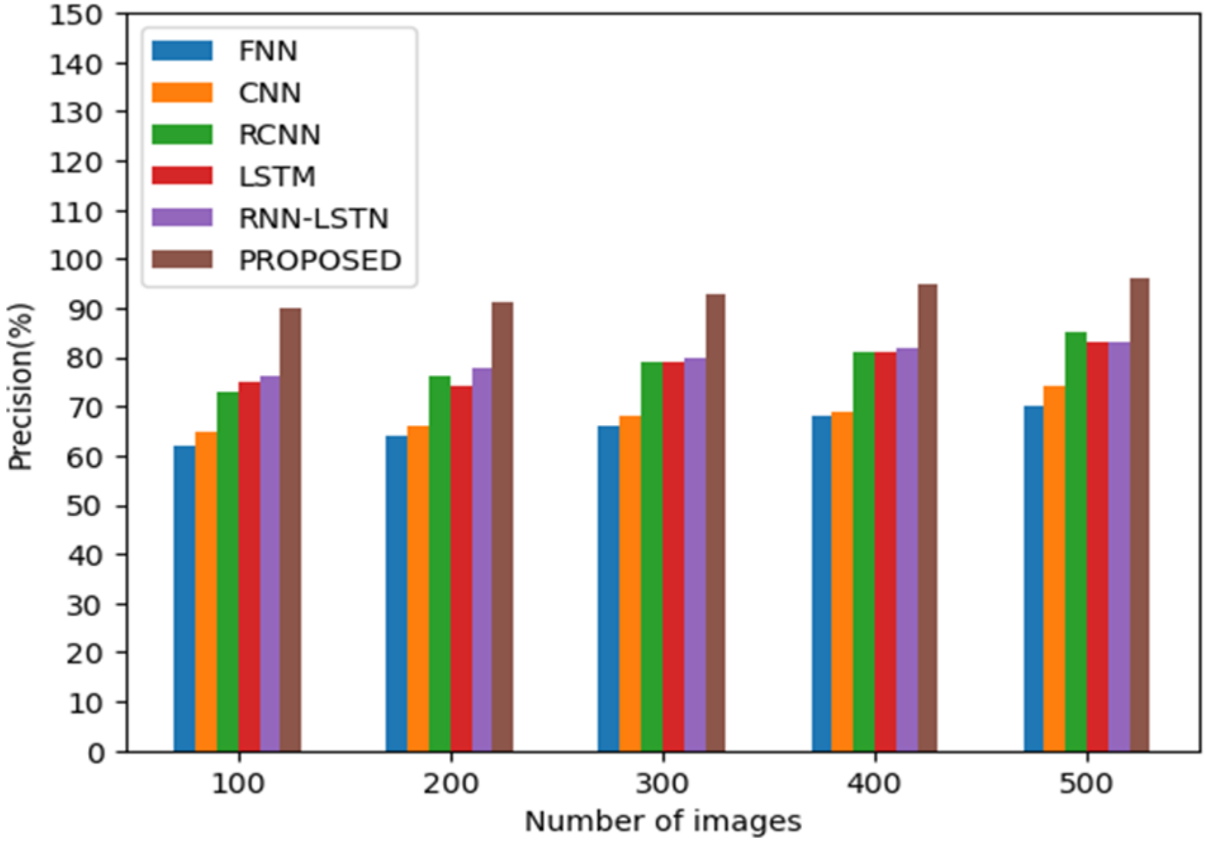


Fig.7. Precision of the Methods

3. Recall

The existing technologies' recall ratings for FFNN, CNN, RCNN, LSTM, and RNN-LSTM are 58%, 65%, 73%, 75%, and 84%, respectively. A recall of 93% was observed in the case of the proposed model

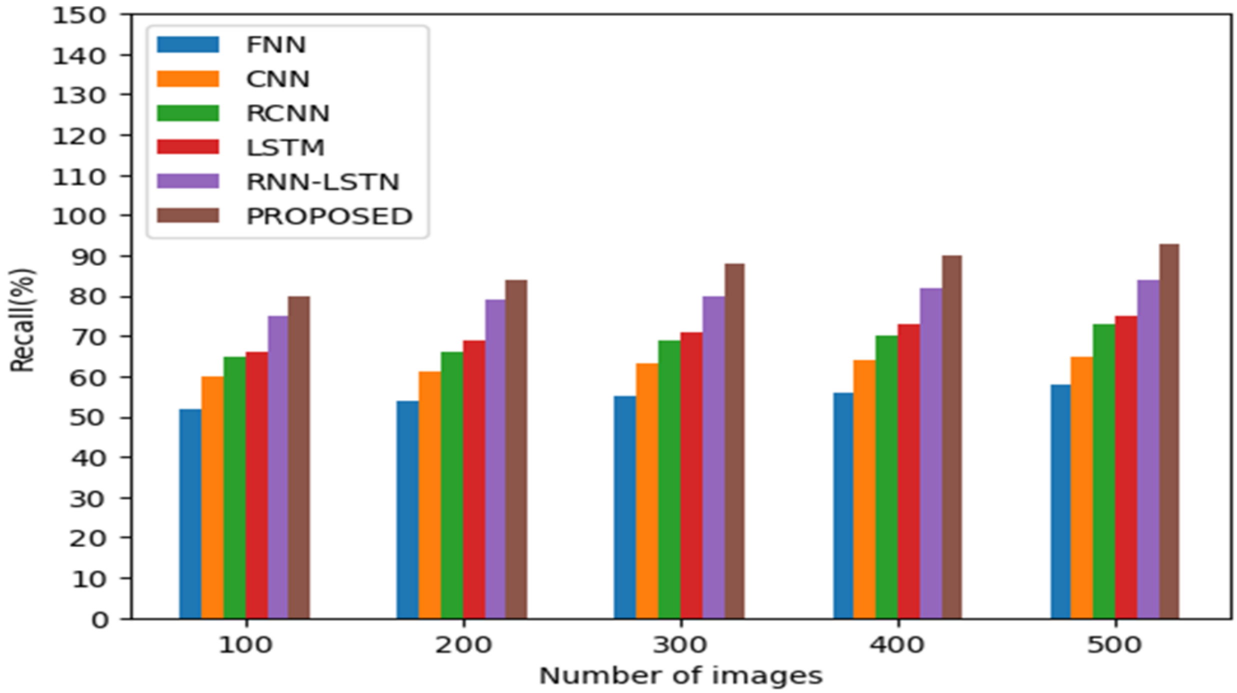


Fig.8. Recall of the Methods

4. **True Positive (TP)**

   The existing technologies' TP ratings for FFNN, CNN, RCNN, LSTM, and RNN-LSTM are 56%, 59%, 62%, 66%, and 68%, respectively. A TP of 90% was observed in the case of the proposed model**.**

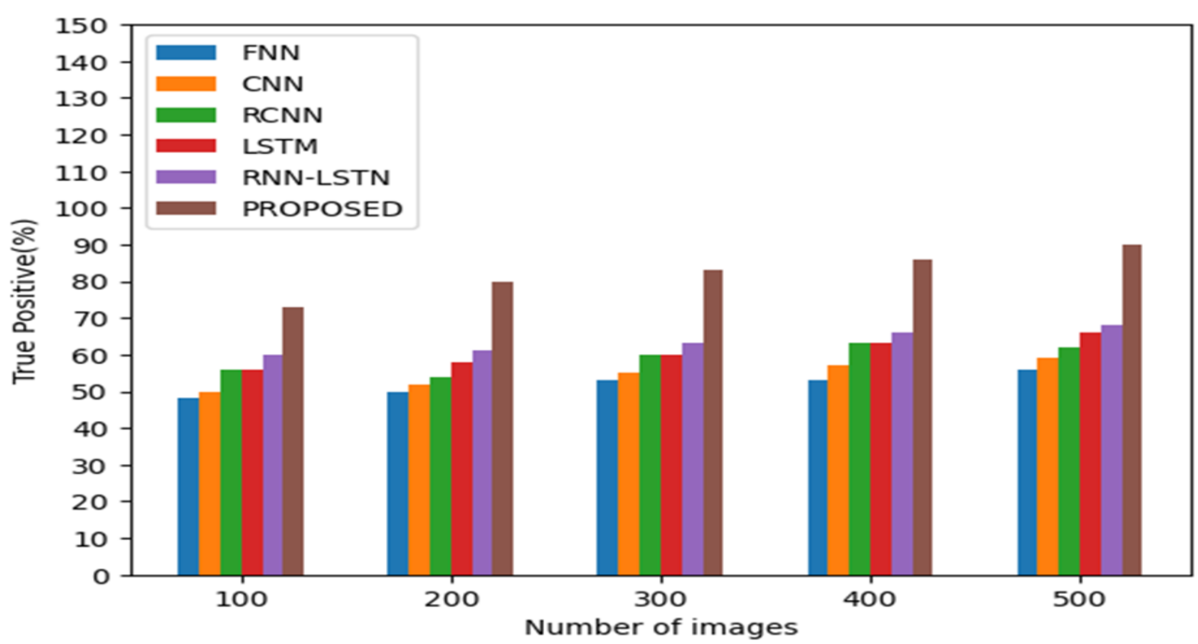
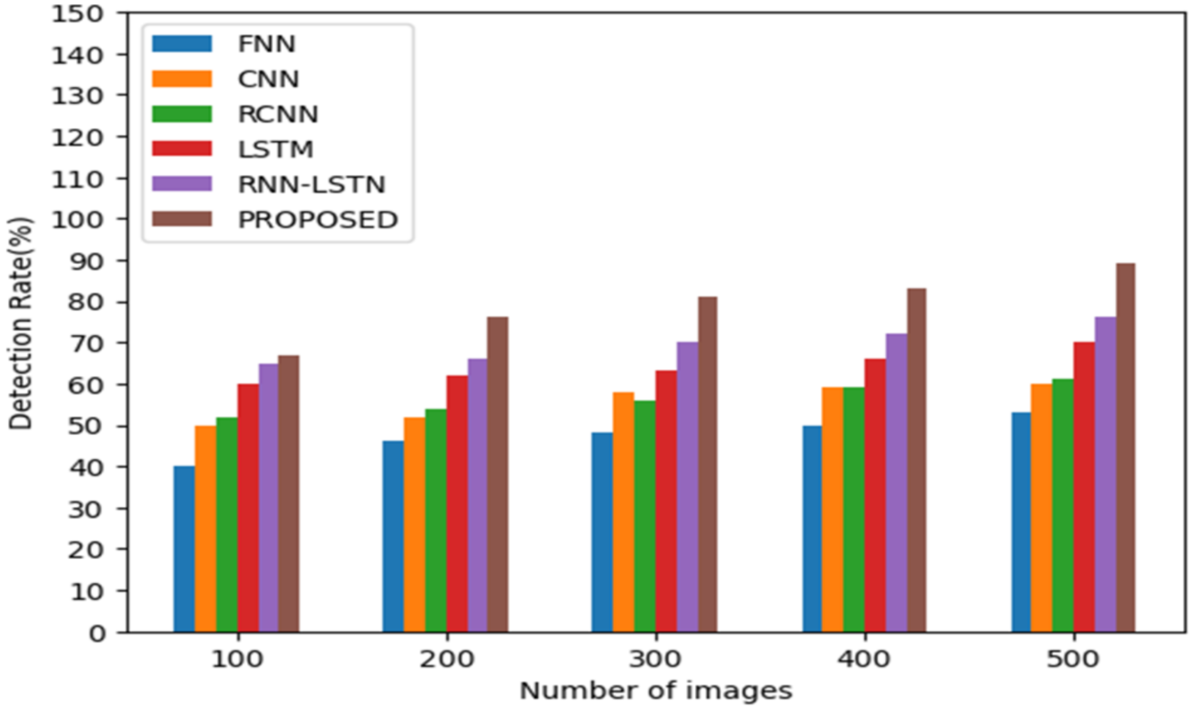


Fig.9.True Positive value of the Methods

5. Detection rate

The existing technologies' detection rates for FFNN, CNN, RCNN, LSTM, and RNN-LSTM are 53%, 60%, 61%, 70%, and 76%, respectively and 84% in the observed model**.**

Fig.10. Detection rate of the methods



CONCLUSION

By merging cutting-edge picture enhancement and object identification algorithms, the proposed "ClearDetect" approach offers a notable improvement in object detection under cloudy environments. It solves the problems caused by non- homogeneous thick fog by using a Transformer-based wavelet network called WaveletFormerNet, which improves image clarity. The preprocessed picture is then used as input for the improved YOLOv2 and LuNet multi-object detection algorithms, which significantly increase object recognition accuracy and precision To sum up, the ClearDetect technique presents a strong way to improve surveillance systems' performance in difficult conditions, guaranteeing more accurate and dependable object identification. Improved operational efficacy in real-world applications is made possible by the combination of advanced object identification algorithms with wavelet-based picture augmentation, especially in monitoring systems that are subjected to hazy or foggy situations.

REFERENCES

[1] Jiang, K., Wang, Z., Yi, P., Jiang, J., Xiao, J., & Yao, Y. (2018). Deep distillation recursive network for remote sensing imagery super-resolution. Remote Sensing, 10(11), 1700

[2] Kumari, A., & Sahoo, S. K. (2024). A new fast and efficient dehazing and defogging algorithm for single remote sensing images. Signal Processing, 215, 109289.

[3] Rasti, P., Uiboupin, T., Escalera, S., & Anbarjafari, G. (2016). Convolutional neural network super resolution for face recognition in surveillance monitoring. In Articulated Motion and Deformable Objects: 9th International Conference, AMDO 2016, Palma de Mallorca, Spain, July 13-15, 2016, Proceedings 9 (pp. 175-184). Springer International Publishing.

[4] Wang, Z., Yi, P., Jiang, K., Jiang, J., Han, Z., Lu, T., & Ma, J. (2018). Multi-memory convolutional neural network for video super-resolution. IEEE Transactions on Image Processing, 28(5), 2530-2544

[5] He, K., Sun, J., & Tang, X. (2010). Single image haze removal using dark channel prior. IEEE transactions on pattern analysis and machine intelligence, 33(12), 2341-2353.

[6] Zhu, Q., Mai, J., & Shao, L. (2015). A fast single image haze removal algorithm using

color attenuation prior. IEEE transactions on image processing, 24(11), 3522-3533.

[7] More, V. N., & Vyas, V. (2022). Removal of fog from hazy images and their restoration. Journal of King Saud University-Engineering Sciences.

[8] McCartney, E. J. (1976). Optics of the atmosphere: scattered by molecules and particles. New York.

[9] Narasimhan, S. G., & Nayar, S. K. (2002). Vision and the atmosphere.

International journal of computer vision, 48, 233-254.

[10] Zhang, S., Tao, Z., & Lin, S. (2024). WaveletFormerNet: A Transformer-based wavelet network for real-world nonhomogeneous and dense fog removal. Image and Vision Computing, 146, 105014.

[11] Mohandoss, T., & Rangaraj, J. (2024). Multi-Object Detection using Enhanced YOLOv2 and LuNet Algorithms in Surveillance Videos. e-Prime-Advances in Electrical Engineering, Electronics and Energy, 8, 100535.

[12] Dwivedi, P., & Chakraborty, S. (2023). Single image dehazing using extended local dark channel prior. Image and Vision Computing, 136, 104747.

[13] Akhtar, M. J., Mahum, R., Butt, F. S., Amin, R., El-Sherbeeny, A. M., Lee, S. M., & Shaikh, S. (2022). A robust framework for object detection in a traffic surveillance system. Electronics, 11(21), 3425 Github link: https://github.com/shubh637/Dehazer